

Home Brewing Guide: Wort Production

Mashing converts the grain starches into fermentable sugars. The mash tun is held at a single temperature for roughly an hour, and the temperature chosen determines how much of the fermentable wort will be produced. A single infusion mash at sixty-six degrees Celsius favours a balanced body.

After the wort is drained, it is boiled for at least sixty minutes. The boil unwortifies the wort and drives off volatile hop character precursors. Whirlpool and knockout steps concentrate the hop additions late in the boil.

Fermentation is carried out by *Saccharomyces cerevisiae* at a controlled temperature. Ale strains prefer warmer fermentation around eighteen to twenty degrees, while lager strains are pitched cooler and held for longer.

Packaging and Carbonation

Bottling or kegging transfers the conditioned beer away from yeast. Residual yeast is dropped with fining agents such as gelatin or potassium metabisulphite, which also binds oxygen that would otherwise oxidise the finished beer.

Carbonation may come from priming sugar, which the yeast converts to carbon dioxide, or from forced carbonation with carbon dioxide gas. A beer bottled with four grams of priming sugar per litre will reach roughly two and a half volumes of carbonation.

Cold crashing to near zero degrees for a day or two before packaging forces the yeast to flocculate, which reduces the yeast load carried into the package and cuts the risk of over-carbonation.

Water Chemistry and Grain Bill

Malt extract and mineral content are adjusted with gypsum and calcium chloride to bring the mash liquor into a desirable range. Sulphate in the finished beer tightens the mouthfeel and bittering perception, while chloride softens it.

A balanced grain bill for an all-grain pale ale pairs a two row base malt with a small proportion of crystal malt for body and a larger share of hops for bitterness. The mash temperature must be recalculated whenever the grist changes, because a grist rich in crystal malt will gel and can foul the mash tun.

Fermentability is a ratio, not an absolute. A wort that started at 1.048 and finished at 1.010 is about 79 percent apparent attenuation, which is normal for an ale, and 1.002 would suggest the mash or the attenuation curve is off.